OSTAAD ERP: OPERATIONAL MANUAL

End-to-End Civil Engineering Lifecycle & Execution Framework

From Project Inception, 15 Pre-Construction Tools, BOQ Takeoff, Dual-Control Procurement,
Site Store / GRN Inward, DPR Progress, to 3-Way Reconciliation and Handover

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System Version: 2.4.0-Enterprise | **Release Date:** March 2026 | **Platform URL:** `https://civix.space`

1 Executive Architecture Overview

Ostaad ERP is a purpose-built, cloud-native Enterprise Resource Planning system tailored for Indian and international infrastructure, commercial real estate, and industrial construction projects. The platform replaces fragmented spreadsheets and disconnected point solutions with a unified, tenant-isolated data backbone across three distinct stages:

1. **Pre-Construction Feasibility & Permitting:** Complete 15-tool analytical suite executing National Building Code (NBC 2016) zoning, irregular land geometry calculations, soil bearing analysis, statutory NOC tracking, and mobilization gating.
2. **During-Construction Execution & Controls:** Master BOQ formulation, dynamic item takeoff using 18 civil engineering precedents, automated material decomposition, dual-control RFQ-to-PO procurement, multi-godown Goods Receipt Note (GRN) inward, double-entry stock ledgers, and Daily Progress Reports (DPR).
3. **Post-Construction Settlement & Closeout:** 3-way automated matching (PO $\leftrightarrow$ GRN $\leftrightarrow$ Tax Invoice), Defect Liability Period (DLP) retention escrow, municipal Occupancy Certificate (OC) signoff, and As-Built drawing archiving.

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Lifecycle Stage	Operating Route	Primary Core Modules
Pre-Construction	`/dashboard/pre-construction`	15 Tools Suite, NBC Setbacks, Soil SBC, Gantt, Cashflow S-Curve.
Project Initiation	`/dashboard/operations`	Tenant entity setup, Client metadata, Project Master initialization.
BOQ Formulation	`/dashboard/operations`	Precedent specifications, WBS Level 1–4, live quantity rate takeoff.
Procurement & RFQ	`/dashboard/operations`	Material decomposition, Comparative Statement (CS), L1 evaluation.
Finance Dual-Control	`/dashboard/finance`	Budget pool hard stops, PO issuance, UTR bank disbursement ledger.
Site Store & GRN	`/dashboard/site-store`	Multi-godowns, weighbridge logging, +5% damage tolerance, stock ledger.
Site Execution	`/dashboard/projects`	DPR, workforce headcount, machinery hours, GFC Drawing Vault.
Post-Construction	`/dashboard/finance`	3-way invoice matching, 5%–10% retention release, OC handover.

Table 1: System Module and Navigation Directory

2 Step 1: Project Setup & Tenant Entity Definition

2.1 Multi-Tenant Operating Entity Switcher

Every construction site operates within a legal corporate entity registered with the Ministry of Corporate Affairs and the Goods and Services Tax (GST) Network. The system header incorporates an active entity selector managing tenant-isolated records:

- **Entity Name:** Legal business name (e.g., *Ostaad Infrastructure & Projects Ltd*).
- **GSTIN Identification:** 15-character alphanumeric GSTIN (e.g., `19AAACO9988P1Z5`).
- **State Jurisdiction:** State code (e.g., Code 19 for West Bengal, Code 27 for Maharashtra).
- **Tenant Data Isolation:** Ensures cross-entity budgets, ledgers, and vendor accounts never bleed into separate contracts.

2.2 Project Inception Modal Workflow

From the Operations dashboard (`/dashboard/operations`), the user clicks **+ Create Project**:

1. **Project Name:** Unique project identifier (e.g., *Sansara Tower A*).
2. **Client / Developer:** Customer name (e.g., *PS Group Consortium*).
3. **Geographical Location:** Site city and district (e.g., *Howrah, West Bengal*).
4. **Project Code Generation:** The system assigns an immutable, unique code (`PRJ-MW6E8SRQ`).
5. **Initialization Options:** Checkbox option to pre-populate standard baseline civil works or start with a blank registry.

3 Step 2: BOQ Formulation & Precedent Takeoff Engine

3.1 The Precedent-Driven BOQ Modal Dialog

Rather than requiring manual spreadsheet preparation, the ERP provides an interactive dialog populated with **18 standardized Indian construction precedent specifications** across 6 core

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engineering divisions:

WBS Code	Category	Engineering Specification Description	UOM	Std. Rate (INR)
1.1.1	Earthwork	Excavation in foundation trenches in ordinary soil up to 1.5m	CUM	285.00
1.1.2	Earthwork	Plinth filling with selected quarry dust in 15cm compacted layers	CUM	420.00
1.1.3	Earthwork	Chemical anti-termite emulsion treatment (IS 6313 compliance)	SQM	110.00
1.2.1	Concrete	Plain Cement Concrete (PCC) 1:4:8 under footings	CUM	4,250.00
1.2.2	Concrete	Reinforced Cement Concrete (RCC) M25 in isolated footings	CUM	6,250.00
1.2.3	Concrete	Reinforced Cement Concrete (RCC) M30 in columns & shear walls	CUM	6,900.00
1.2.4	Concrete	Ready-Mix Concrete (RMC) M35 pumpable mix for post-tensioned slabs	CUM	7,450.00
2.1.1	Steel	Thermo-Mechanically Treated (TMT) Fe-550D rebar (8mm–32mm)	MT	58,500.00
2.1.2	Steel	Structural steel fabrication and erection in trusses and bracing	MT	74,000.00
2.1.3	Steel	18-gauge annealed mild steel binding wire for intersections	KG	85.00
3.1.1	Masonry	Autoclaved Aerated Concrete (AAC) blocks 200mm with polymer mortar	CUM	4,850.00
3.1.2	Masonry	Fly ash brick masonry 230mm thick in cement mortar 1:6	CUM	5,200.00
3.1.3	Masonry	Half-brick internal partition wall 115mm with hoop reinforcement	SQM	780.00
3.2.1	Finishes	Internal cement plaster 12mm thick in cement mortar 1:4	SQM	240.00
4.1.1	Finishes	Double-charged vitrified floor tiles 800×800mm with adhesive	SQM	1,150.00
4.2.1	Finishes	Premium interior acrylic emulsion paint (2 coats over primer)	SQM	165.00
5.1.1	Waterproofing	Two-component polymer elastomeric coating in wet areas	SQM	520.00
5.2.1	MEP	CPVC SDR-11 water supply piping concealed with brass fittings	RMT	340.00

Table 2: Standard Indian Civil Engineering Precedents Library

3.2 Manual / Custom Mode & Live Amount Calculation

When the user selects `-- None / Custom Specification (Manual Entry) --`, all fields unlock:

- The user defines a custom WBS code (e.g., `GEN-401`) and custom text descriptions.
- Real-time frontend math calculates total committed cost:

$$\text{Total Line Amount (INR)} = \text{Quantity} \times \text{Unit Rate (INR)} \tag{1}$$

- Line items enter the register with status `In Review` (rendered on a strict single line with zero text wrapping).
- Clicking **Finalise BOQ** validates all lines, changes statuses to `Finalised`, and unlocks downstream material decomposition.

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4 Step 3: The 15 Pre-Construction Planning & Feasibility Suite

The Pre-Construction Engine (`/dashboard/pre-construction`) features 15 integrated analytical tools divided into three engineering phases:

4.1 Phase A: Site Viability, Regulatory Zoning & Preliminary Finance

1. **Tool 1: Site Survey GO / NO-GO Verification Engine**
   Executes a 10-point weighted inspection verifying geotechnical access, stormwater flooding risks, overhead high-tension (HT) electrical lines, proximity to railway tracks/waterbodies, and water table depth.

$$\text{Survey Viability Score (\%)} = \frac{\sum_{i=1}^{10} \text{Weight}_i \times \text{Pass}_i}{\sum_{i=1}^{10} \text{Weight}_i} \times 100 \quad (\geq 75\% = \text{GO}) \tag{2}$$

2. **Tool 6: Plot Area, Irregular Geometry & Setbacks Calculator**
   Computes precise plot acreage from irregular polygon survey coordinates. Converts units across Metric, British, and Indian regional land measures:

$$1 \text{ Acre} = 43,560 \text{ sq.ft} = 4,046.86 \text{ sq.m} = 40 \text{ Gunthas} = 3.025 \text{ Bighas (Bengal)} \tag{3}$$

3. **Tool 12: Zoning, Setbacks & Permissible FAR Engine (NBC 2016)**
   Applies National Building Code norms according to access road width:

$$\text{Max Built-Up Area (BUA)} = \text{Plot Area} \times \text{Base FAR} + \text{Purchasable Premium FAR} \tag{4}$$

   Determines mandatory front, rear, and lateral clearances based on building height H:

$$\text{Front Setback (m)} = \max\left(6.0,\ \frac{H - 15}{3} + 4.5\right) \tag{5}$$

4. **Tool 8: Geotechnical Soil Stratification & Safe Bearing Capacity (SBC)**
   Analyzes Standard Penetration Test (SPT N-values) across strata depths:

$$\text{Net Allowable Bearing Pressure } q_a = 12.5\, N \left(1 + \frac{0.3}{B}\right)^2 R_\gamma \quad (\text{in kN/m}^2) \tag{6}$$

   Provides structural foundation advisory: Isolated Footings ($\geq$ 200 kN/m^2), Raft Foundation (100–180 kN/m^2), or Bored Cast-in-Situ Piling ($<$ 100 kN/m^2).

5. **Tool 2: Pre-Construction Cost Estimator & CPWD DSR Rates**
   Translates architectural massing into preliminary elemental budgets based on Central Public Works Department (CPWD) Delhi Schedule of Rates (DSR).

6. **Tool 9: Financial Feasibility, Debt/EMI Modeling & DSCR**
   Evaluates project capital structuring, builder equity requirements, and loan amortizations:

$$\text{Debt Service Coverage Ratio (DSCR)} = \frac{\text{Net Operating Income (NOI)}}{\text{Annual Principal + Interest Service}} \quad (\text{Target } \geq 1.35) \tag{7}$$

7. **Tool 14: Options Gate Matrix (IS 13920 Structural Framing)**
   Multi-attribute comparison evaluating Cast-in-Situ RCC Frame vs. Post-Tensioned Slabs vs. Precast Concrete vs. Structural Steel on cost, crane dependency, and seismic ductilization.

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4.2 Phase B: Estimation, Scheduling & Statutory Permitting

8. **Tool 3: Master BOQ Importer & WBS Codifier**
   Bulk Excel/CSV takeoff ingestion validating standard WBS coding structures (Level-1 Phase, Level-2 Package, Level-3 Work Item, Level-4 Task).

9. **Tool 7: Historic Vendor Rate Benchmarking**
   Tracks variance between current market bids against historical rates indexed over the preceding 24 months ($L1$ vs. $V1$ benchmark).

10. **Tool 4: Gantt & Critical Path Method (CPM) Scheduler**
    Calculates Early Start (ES), Late Start (LS), Early Finish (EF), and Late Finish (LF) across all dependencies. Pinpoints zero-float activities driving the critical path:

$$\text{Total Float } TF = LS - ES = LF - EF \quad (TF = 0 \implies \text{Critical Path}) \tag{8}$$

11. **Tool 10: Cashflow S-Curve Forecast & Monthly Burn Demand**
    Generates cumulative capital expenditure projections modeled via Sigmoid / Cumulative Normal Distribution:

$$S(t) = \frac{\text{Approved Capex}}{1 + e^{-k(t-t_0)}} \tag{9}$$

12. **Tool 5: Statutory NOCs & Municipal Permitting Pipeline**
    Real-time tracking of mandatory legal clearances: Title Search Clearance, Airport Authority of India (AAI) Obstacle Clearance, Fire Department Provisional NOC, State Pollution Control Board Consent to Establish (CTE), and RERA Registration.

4.3 Phase C: Site Mobilization Readiness & Document Governance

13. **Tool 11: Site Logistics, Laydown & Equipment Layout**
    Dynamic zone allocation for 50-tonne mobile cranes, batching plants, rebar fabrication tables, aggregate bins, and security cabins.

14. **Tool 13: Budget Variance & Contingency Risk Allocation**
    Calculates dynamic contingency reserves (5%–10%) based on site soil risks and municipal permit delays.

15. **Tool 15: 50-Point Ground Mobilization Readiness Gate**
    Rigorous pre-construction audit covering boundary barricading, CCTV installation, labor camp EHS compliance, water/power connections, and builder risk insurance. Requires 100% signoff before earthmoving commences.

16. **Cloud Project Document Vault & GFC Drawing Engine**
    Repository stamping and archiving Good-For-Construction drawings (`Rev-0`, `Rev-1`, `Rev-2`) with immutable cryptographic audit tracking.

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5 Step 4: During Construction (Execution, Procurement, Store & Finance)

5.1 Procurement Decomposition & RFQ Generation

Once the BOQ is finalized, the system automatically decomposes line items into discrete procurement material lots:

- Concrete works decompose into **OPC 53 Cement Bags**, **Crushed Aggregates (10mm/20mm)**, and **M-Sand**.
- Rebar line items decompose into diameter-specific bundles (**8mm, 10mm, 12mm, 16mm, 20mm, 25mm, 32mm**).
- RFQs are generated and transmitted to verified, GST-registered suppliers.

5.2 Vendor Comparative Statement (CS) Matrix

Incoming supplier quotations are parsed side-by-side:

$$\text{Landed Cost} = \text{Base Price} + \text{Freight / Loading} + \text{CGST}(9\%) + \text{SGST}(9\%) - \text{Input Tax Credit (ITC)} \tag{10}$$

The Project Manager submits an $L1$ selection justification to the Commercial Directorate.

5.3 Finance Directorate Dual-Control PO Approval

Commercial safety is enforced via automated budget headroom checks:

1. **Hard-Stop Budget Pools:** Dedicated budget reserves exist for Civil Structure, Steel Rebar, Cement, Sand/Aggregates, and Finishes.
2. If Committed Spend + Pending PO > Approved Headroom, the system halts PO issuance.
3. Upon CFO approval, an authorized Purchase Order (PO) PDF is minted with contractual payment milestones (e.g., *15% Advance against Bank Guarantee, 85% against Gate Weighbridge GRN*).
4. Bank disbursements are logged with Unique Transaction Reference (UTR) numbers.

5.4 Site Store Operations, Multi-Godown & GRN Inward

When material arrives at the project gate:

1. **Weighbridge & Vehicle Logging:** Vehicle registration, driver contact, and gross tare weight are recorded.
2. **Goods Receipt Note (GRN):** Storekeepers inspect cargo and issue a digital GRN.
3. **Damage Tolerance Gate:** Damaged cement bags or corroded steel beyond the **+5% tolerance** threshold trigger an automated vendor debit note and partial rejection.
4. **Double-Entry Stock Ledger:** Material shifts from *In-Transit* to *Physical Godown Stock* (Central Cement Shed, Rebar Laydown Yard, Aggregate Silos).
5. **Material Indent Slips:** Subcontractors can only withdraw materials against active WBS tasks, preventing site pilferage and tracking wastage percentages.

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5.5 Daily Progress Reporting (DPR) & Engineering Vault

Site operations are updated daily in `/dashboard/projects`:

- **Workforce Headcount:** Trade-specific counts (Bar-benders, Masons, Carpenters, Welders, Helpers).
- **Plant & Machinery Log:** Operational vs. idle hours for Tower Cranes, Concrete Boom Pumps, Transit Mixers, and Excavators.
- **Activity Progress:** Structural pours completed, linear meters plastered, and hindrances encountered.
- **GFC Stamping:** Site engineers execute exclusively from *Good For Construction* stamped drawing sheets.

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6 Step 5: Post-Construction, Reconciliation & Final Closure

6.1 Automated 3-Way Invoice Reconciliation

Prior to disbursing vendor or contractor final payments, the finance engine executes an automated 3-way audit:

$$\underset{\text{(Authorized Quantity \& Rate)}}{\textbf{Purchase Order (PO)}} \longleftrightarrow \underset{\text{(Gate Accepted Inward Weight)}}{\textbf{Goods Receipt Note (GRN)}} \longleftrightarrow \underset{\text{(Billed Amount \& GSTIN)}}{\textbf{Vendor Tax Invoice}} \tag{11}$$

Any variance in quantities (GRN $<$ Invoice) or unapproved rate escalations (PO Rate $<$ Billed Rate) is highlighted, holding payment pending credit notes.

6.2 Defect Liability Period (DLP) & Retention Money Management

To safeguard structural integrity and finish quality:

- The system deducts **5% to 10% Retention Money** from all progressive Running Account (RA) bills.
- Retention funds are placed in a contractually designated escrow ledger.
- During the 12-to-24 month Defect Liability Period (DLP), any structural seepage, plaster cracking, or mechanical failures are remedied under contractor warranty.
- Upon joint inspection signoff by the Lead Resident Engineer and Quality Head, retention funds are released.

6.3 Statutory Handover & Municipal Occupancy Certificate (OC)

The project transitions to final commissioning:

1. **Fire Safety Final NOC:** Inspection of wet risers, sprinkler heads, smoke dampers, and fire tender turning radii.
2. **Environmental Audit:** Final Consent to Operate (CTO) issued for the Sewage Treatment Plant (STP) and Organic Waste Converter (OWC).
3. **Occupancy Certificate (OC):** Municipal development authority conducts the final physical survey and issues the formal OC.

6.4 As-Built Archiving & Executive Governance Closeout

- Final **As-Built CAD/PDF drawings** are permanently locked in the Cloud Project Vault.
- Final earned value metrics are archived for executive audit:

$$\text{Cost Performance Index (CPI)} = \frac{\text{Earned Value (EV)}}{\text{Actual Cost (AC)}} \tag{12}$$

$$\text{Schedule Performance Index (SPI)} = \frac{\text{Earned Value (EV)}}{\text{Planned Value (PV)}} \tag{13}$$

- Vendor No-Dues Certificates (NDCs) are captured, and the project status transitions from `Active Execution` to `Handed Over / Closed`.

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End of Operational Lifecycle Specification

Compiled and Verified for Ostaad Construction ERP Suite

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